Oil and Airline ETF Dataset Appendix

1.1: Dataset Overview

This project uses publicly available financial market data to examine whether oil price changes help forecast airline ETF returns at the monthly horizon. The dataset contains price and return information for three securities:

- CL=F : WTI crude oil futures
- JETS : U.S. Global Jets ETF
- SPY : SPDR S&P 500 ETF

The raw data were downloaded from Yahoo Finance using the 'yfinance' Python package. Daily adjusted closing prices were extracted and then transformed into monthly price and return series. A final forecasting dataset was constructed by engineering lagged return predictors, rolling volatility measures, and a next-month target variable for JETS returns.

This appendix documents both the initial raw market dataset and the final analyzed forecasting dataset. It also summarizes the key exploratory figures used to understand the data before forecasting models were estimated.

1.2: Unit of Observation

There are two relevant units of observation in this project

Raw dataset:

Each row in the raw dataset represents **one trading day** in the Yahoo Finance download.

- Observational unit = single trading day
- Each observation contains daily market fields for the selected securities.

Final analyzed dataset:

Each row in the final analyzed dataset represents one month-end observation after daily price data were converted into monthly returns and forecasting variables were created.

- Observational unit = single month-end observation
- Each observation contains current monthly returns, lagged predictors, rolling volatility measures, and the next-month JETS return target

1.3: Dataset Construction and Processing

The final modeling dataset was created from the raw daily market data in the following steps:

1. Download raw daily market data for CL=F, JETS, and SPY from Yahoo FInance.
2. Save the download raw file exactly as retrieved for archival and reproducibility.
3. Extract adjusted close prices for the three securities
4. Restrict the working sample to dates where all three series are available.
5. Resample daily close prices to month-end prices.
6. Compute monthly simple returns using percentage changes.
7. Create lagged oil, market and JETS returns features.
8. Create rolling 3-month volatility measures
9. Define the target variable as next-month JETS return.

A key point in this project is that JETS begins later than CL=F and SPY, so the common working sample starts after the first valid JETS observation. This is why the final analysis sample begins later than the raw Yahoo Finance download window.

1.4: Variable Documentation

Varibles in the Raw daily market dataset:

1. Variable: Date
   - Type: Datetime
   - Description: Trading date of the market observation.

2. Variable: Ticker
   - Type: Categorical (string)
   - Description: Identifies the security associated with the observation.
   - Possible Values: CL=F, JETS, SPY

3. Variable: Close
   - Type: Continuous numeric
   - Description: Adjusted closing price of the security. This is the primary price series used to construct monthly prices and returns.

4. Variable: Open
   - Type: Continuous numeric
   - Description: Opening price of the security for the trading day.

5. Variable: High

- Type: Continuous numeric
- Description: Highest intraday price of the security for the trading day.

6. Variable: Low
   - Type: Continuous numeric
   - Description: Lowest intraday price of the security for the trading day.

7. Variable: Volume
   - Type: Integer / continuous numeric
   - Description: Daily trading volume of the security.

Variables in the final analyzed monthly forecasting dataset

8. Variable: CL=F
   - Type: Continuous nu
   - Description: Monthly return of WTI crude oil futures.
   - Interpretation: Positive values indicate oil appreciated during the month; negative values indicate oil declined.

9. Variable: JETS
   - Type: Continuous numeric
   - Description: Monthly return of the airline ETF.
   - Interpretation: This is the sector return being studied.

10. Variable: SPY
    - Type: Continuous numeric
    - Description: Monthly return of the S&P 500 ETF.
    - Interpretation: Used as a broad market benchmark and market control variable.

11. Variable: oil_lag_1
    - Type: Continuous numeric
    - Description: Oil return from one month prior.

12. Variable: oil_lag_2
    - Type: Continuous numeric
    - Description: Oil return from two months prior.

13. Variable: oil_lag_3
    - Type: Continuous numeric
    - Description: Oil return from three months prior.

14. Variable: oil_lag_6
    - Type: Continuous numeric

- Description: Oil return from six months prior.

15. Variable: spy_lag_1
    - Type: Continuous numeric
    - Description: SPY return from one month prior.

16. Variable: spy_lag_2
    - Type: Continuous numeric
    - Description: SPY return from two months prior.

17. Variable: spy_lag_3
    - Type: Continuous numeric
    - Description: SPY return from three months prior.

18. Variable: jets_lag_1
    - Type: Continuous numeric
    - Description: JETS return from one month prior.

19. Variable: jets_lag_2
    - Type: Continuous numeric
    - Description: JETS return from two months prior.

20. Variable: jets_lag_3
    - Type: Continuous numeric
    - Description: JETS return from three months prior.

21. Variable: jets_lag_6
    - Type: Continuous numeric
    - Description: JETS return from six months prior.

22. Variable: oil_vol_3
    - Type: Continuous numeric
    - Description: Rolling 3-month standard deviation of oil returns.
    - Interpretation: Measures recent oil volatility.

23. Variable: spy_vol_3
    - Type: Continuous numeric
    - Description: Rolling 3-month standard deviation of SPY returns.
    - Interpretation: Measures recent market volatility.

24. Variable: jets_vol_3
    - Type: Continuous numeric
    - Description: Rolling 3-month standard deviation of JETS returns.
    - Interpretation: Measures recent airline ETF volatility.

25. Variable: target_next_jets
    - Type: Continuous numeric
    - Description: Next-month JETS return.
    - Interpretation: This is the target variable used in the forecasting analysis.

1.5: Descriptive Statistics

The monthly return data show several important patterns that motivate the forecasting exercise.

- **Oil returns are the most volatile series in the sample.**
- **JETS returns are more volatile than SPY returns.**
- **SPY has the highest average monthly return over the sample.**
- **JETS and SPY move together more closely than JETS and oil.**

These descriptive findings suggest that oil is economically relevant but also noisy, while JETS is influenced substantially by broader market movements. This is important because the project is not simply asking whether oil matters in theory, but whether oil improves actual out-of-sample forecasts of airline ETF returns.

1.6: Figures and Visualization

1. **<u>Normalized Monthly Prices(Line Plot)</u>**

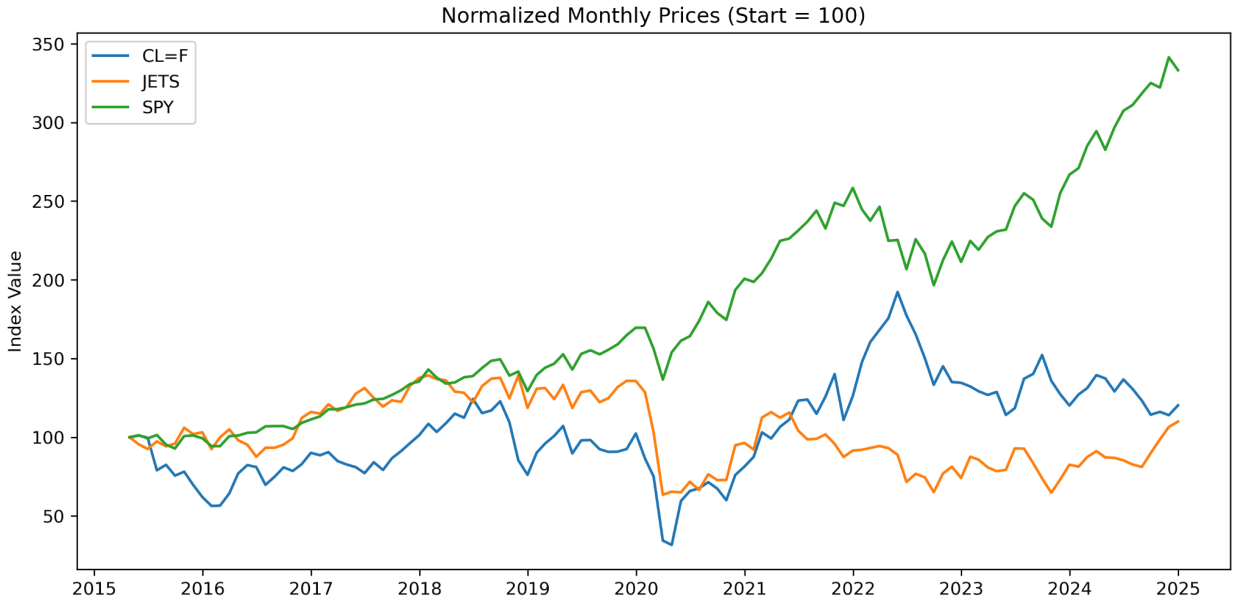

Figure Description:

This figure compares normalized month-end price paths for CL=F, JETS, and SPY, with all series rebased to 100 at the beginning of the common sample.

Variables Used:

- CL=F
- JETS
- SPY

What the Figure Shows:

- SPY rises the most over the sample period
- JETS underperforms SPY over time
- OIL prices show stronger cyclical movement and instability.

Interpretation:

The three assets follow very different long-run trajectories. This supports the decision to model returns rather than raw prices, since raw prices are not directly comparable across assets with different scales and long-run trends.

2. **Monthly Returns (Line Plot)**

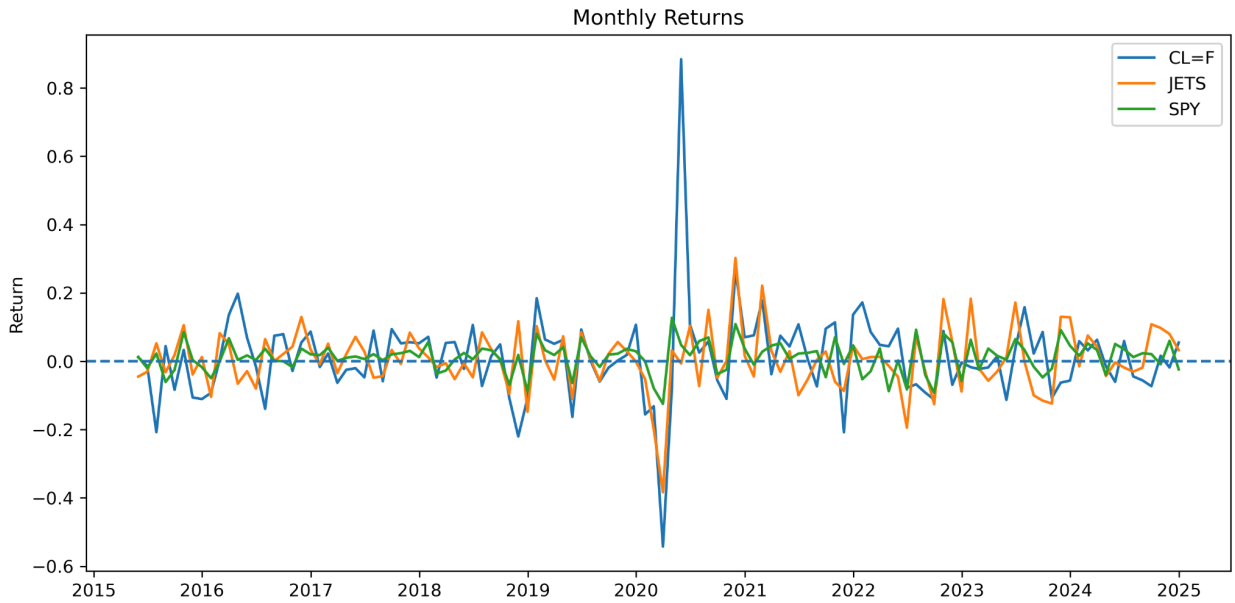

Figure Description:

This figure plots monthly returns for CL=F, JETS, and SPY.

Variables Used:

- CL=F
- JETS
- SPY

What the Figure Shows:

- Oil returns fluctuate much more sharply than JETS and SPY
- JETS and SPY often move together in the same direction
- Large market disturbances are visible, especially around the 2020 period.

Interpretation:

Oil may be an informative economic variable, but it is also highly volatile and unstable. JETS appears more visually linked to the broad market than to oil alone, which motivates the inclusion of SPY as a control and benchmark predictor.

3. **Correlation Matrix of Monthly Returns**

Figure Description:

This figure summarizes pairwise correlation among monthly returns for CL=F, JETS, and SPY.

Variables Used:

- CL=F
- JETS
- SPY

What the Figure Shows:

- JETS is more strongly correlated with SPY than with oil.
- Oil and SPY also show a positive relationship, but a weaker one.

Interpretation:

This is one of the most important exploratory findings in the project. It suggests that airline ETF returns may be driven more by general market conditions than by oil changes alone. As a result, the research question shifts from “Does oil matter?” to “Does oil add forecasting value beyond what the market already explains?”

4. **<u>12- Month Rolling Correlation: Oil vs JETS</u>**

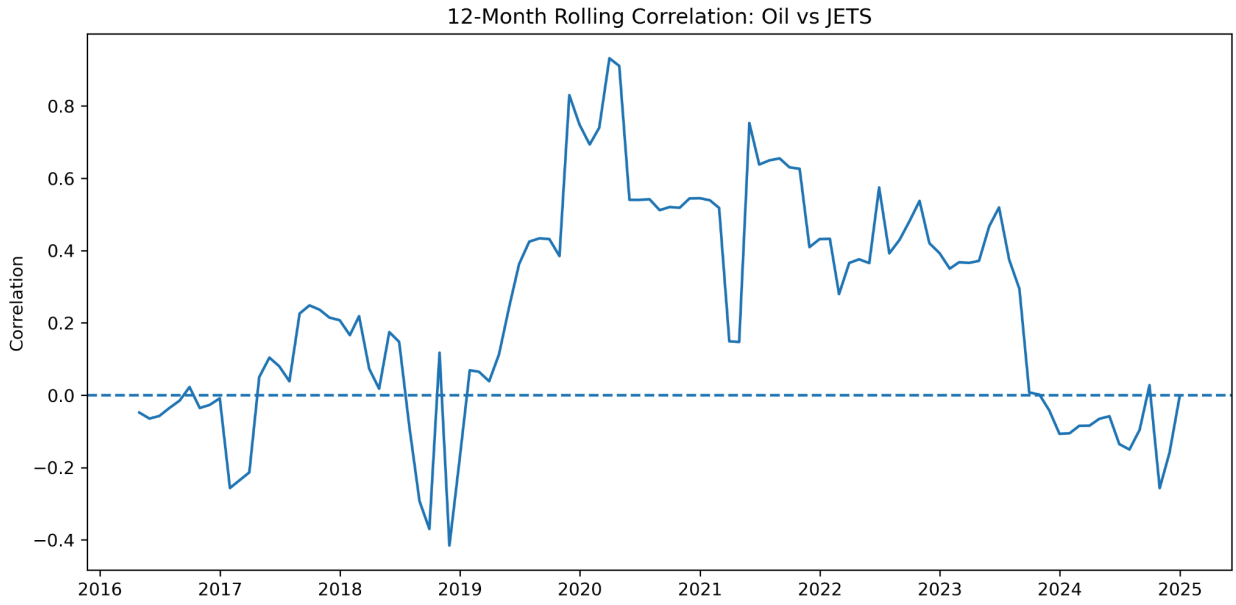

Figure Description:

This figure plots the rolling 12-month correlation between oil returns and JETS returns.

Variables Used:

- CL=F
- JETS

What the Figure Shows:

- The relationship between oil and JETS changes over time
- The oil-JETS correlation is not stable across samples.

Interpretation:

The oil and airline relationship appears time-varying rather than constant. This means that even if oil helps explain airline returns in some periods, that relationship may weaken or reverse in others.

5. **<u>Focused Model Comparison by RMSE</u>**

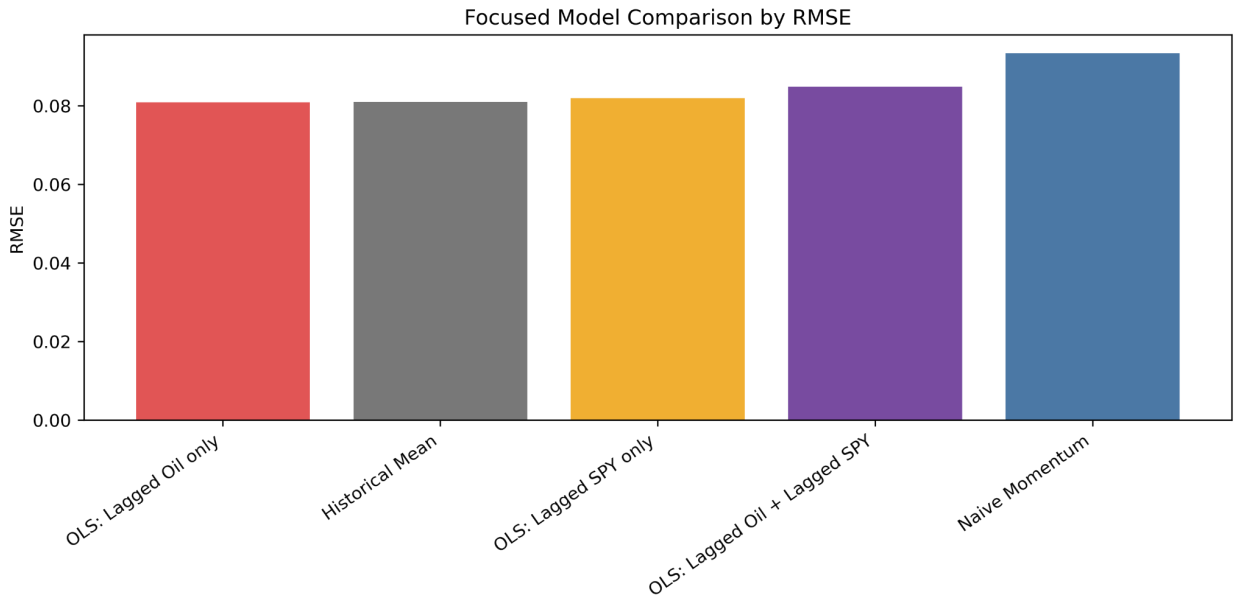

Figure Description:

This bar chart compares a reduced set of representative forecasting models using RMSE.

Variables Used:

- Model name
- RMSE

Models shown:

- Historical Mean
- Naive Momentum
- OLS: Lagged SPY only
- OLS: Lagged Oil only
- OILS: Lagged Oil + Lagged SPY

What the Figure Shows:

- The lagged oil-only OLS model has the lowest RMSE
- Historical Mean performs almost as well.
- The combined Oil + SPY model does not materially improve on the oil-only model.

Interpretation:

Lagged oil returns may contain a small amount of predictive information for next-month JETS returns. However, the improvement over simple benchmark models is very small. The main conclusion is not that the oil model is powerful, but that no model provides a strong forecasting edge.

6. Focused Model Comparison by Directional Accuracy

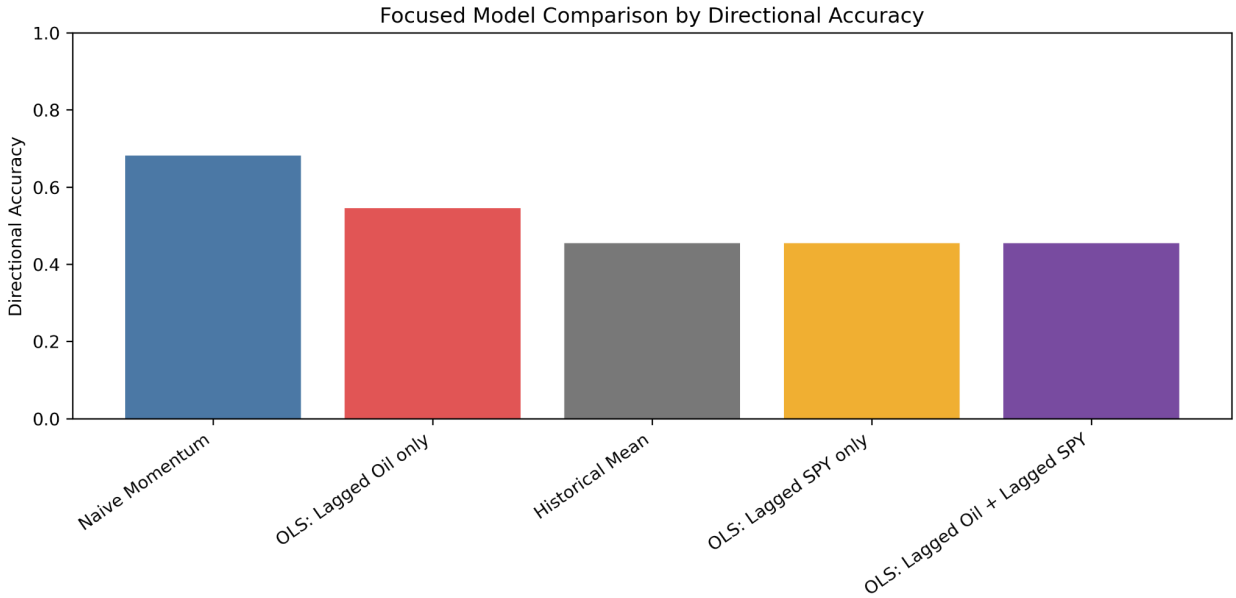

Figure Description:

This bar chart compares the same representative models using directional accuracy.

Variables Used:

- Model name
- RMSE

What the Figure Shows:

- Naive momentum performs best at forecasting the sign of the next-month JETS return.
- Some models do better at predicting direction than magnitude.

Interpretation:

Directional forecasting and magnitude forecasting are not the same problem. A model can be better at identifying whether JETS will go up or down while still performing poorly in terms of RMSE. In this project, momentum appears more helpful for direction, while lagged oil-only OLS performs slightly better on point-forecast error.

7. Actual vs Predicted Next-Month JETS Returns

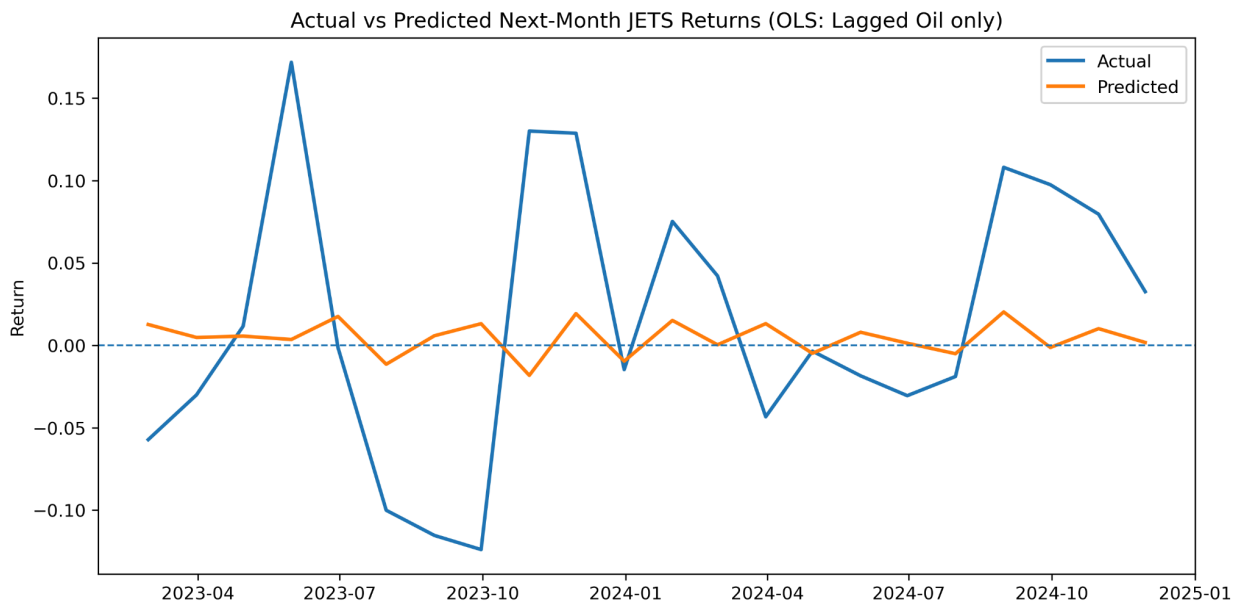

Figure Description:

This figure compares actual next-month JETS returns with the predicted value from the best RMSE model.

Variables Used:

- target_next_jets
- Predicted next-month JETS return

What the Figure Shows:

- Predicted returns remain close to zero most of the time
- The model misses many of the larger positive and negative swings in realized returns.

Interpretation:

Even the best model in the comparison struggles to capture the magnitude of large moves. This reinforces the main conclusion of the project: the predictive signal is weak in practical out-of-sample forecasting terms.

1.7: Statistical Procedures

1. **Chronological Train-Test Split**

Purpose:

To evaluate models in a realistic forecasting setting.

Procedure:

The final modeling dataset was divided chronologically into an 80% training sample and a 20% test sample.

Conclusion:

Models were evaluated on future observations rather than on in-sample fit, making the results more realistic and less biased.

2. **Root Mean Squared Error (RMSE)**

Purpose:

To measure the average size of forecast errors, with larger errors penalized more heavily.

Interpretation:

Lower RMSE indicates better point-forecast performance.

Conclusion:

The lagged oil-only OLS model had the lowest RMSE, but only by a very small margin over simple benchmark models.

3. **Mean Absolute Error (MAE)**

Purpose:

To measure the average absolute size of forecast errors.

Interpretation:

Lower MAE indicates more accurate predictions on average.

Conclusion:

MAE results are broadly consistent with RMSE: no model substantially outperforms the basic benchmarks.

4. **Correlation Between Actual and Predicted Returns**

Purpose:

To assess how closely model predictions move with realized returns.

Interpretation:

Higher positive values indicate stronger alignment between predicted and actual return movements.

Conclusion:

Most model exhibit weak actual predicted correlation, indicating limited predictive power

5. **Directional Accuracy**

Purpose:

To measure how often a model correctly predicts the sign of next-month JETS returns.

Interpretation:

Higher directional accuracy indicates better success in predicting whether the ETF rises or falls.

Conclusion:

Naive momentum performs best on directional accuracy, suggesting that recent JETS return persistence may help with sign prediction even when it does not improve RMSE.

1.8: Data Limitations

- The common sample begins in 2015 because JETS data are not available for the full raw download period.
- Monthly data produce a relatively small sample size for forecasting
- Airline ETF returns are influenced by many factors beyond oil prices, including macroeconomic conditions, interest rates, demand shocks, and market sentiment.
- The common sample begins in 2015 because JETS data are not available for the full raw download period.
- Because the best RMSE model predicts values close to zero, even the strongest-performing specification has limited practical forecasting usefulness.

1.9: Appendix Summary

This appendix documents the raw Yahoo Finance market data, the transformation of that data into a monthly forecasting dataset, and the key exploratory figures used in the analysis. The final analyzed dataset is a model-ready monthly panel of returns, lagged predictors, volatility measures, and a next-month JETS return target. The exploratory evidence suggests that oil is economically relevant but highly volatile, while JETS is more strongly linked to the broad market than to oil alone. The forecasting results indicate that lagged oil returns may provide a small amount of information, but the overall out-of-sample predictive signal is weak.